

Splunk Observability Cloud

Collector Components

Receivers, processors, exporters, extensions, and connectors of the Splunk Distribution of the OpenTelemetry Collector

Five Component Types

You activate components by configuring service pipelines in the Collector config

Receivers

Get data into the Collector from multiple sources

Processors

Perform operations on data before it's exported (for example, filtering)

Exporters

Send data to one or more backends or destinations

Extensions

Extend the capabilities of the Collector

Connectors

Send telemetry data between different Collector pipelines

Receivers

Get data into the Collector — over 50 supported sources

Examples by pipeline type:

Metrics

- Host metrics
- Prometheus
- MySQL / PostgreSQL
- vCenter
- StatsD
- Kafka metrics

Logs

- Filelog
- Fluent Forward
- Journald
- Syslog
- TCP log / UDP log
- Windows event log

Traces

- OTLP
- Jaeger
- Zipkin
- Kafka

Kubernetes & Cloud

- Kubelet stats
- Kubernetes cluster
- AWS ECS container metrics
- Azure event hub
- Cloud Foundry

Processors

Transform, filter, or enrich data between receiving and exporting

Batch

Groups data to reduce the number of outgoing requests

Resource / Resource detection

Sets or auto-detects resource attributes, e.g. host or cloud metadata

Memory Limiter

Prevents out-of-memory situations by shedding load

Redaction

Removes sensitive data from telemetry before export

Attributes

Adds, updates, or deletes attributes on spans, metrics, or logs

Filter

Drops telemetry that matches specified conditions

Tail sampling

Samples traces based on criteria evaluated after the full trace is seen

Transform

Modifies telemetry using the OpenTelemetry Transformation Language (OTTL)

Exporters

Send processed data to one or more backends

SignalFx

Sends metrics and traces to Splunk Infrastructure Monitoring / APM

Splunk HEC

Sends logs and events to Splunk Enterprise or Splunk Cloud Platform via HEC

Kafka

Publishes telemetry data to a Kafka topic

AWS S3

Writes telemetry data to an Amazon S3 bucket

OTLP / OTLP-HTTP

Sends data using the native OpenTelemetry protocol

Debug

Writes telemetry to the Collector's own logs — useful for troubleshooting

File

Writes telemetry data to a local file

Load balancing

Distributes data across multiple backend Collector instances

Extensions & Connectors

Support capabilities and link pipelines together

Extensions

Extend Collector capabilities outside the main data pipelines.

Health Check — Exposes an HTTP endpoint to report Collector health

zPages — Serves live in-process diagnostic pages

OAuth2 Client Auth / Basicauth / Bearer — Handle authentication for other components

Docker / Host / Kubernetes observer — Discover targets for automatic discovery

File storage — Persists queue and checkpoint data to disk

Connectors

Join two pipelines, consuming data as an exporter and emitting it as a receiver.

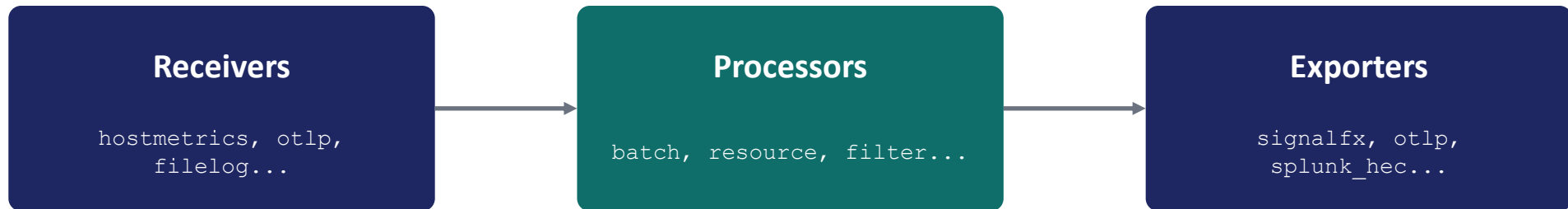
Routing connector — Routes data to different pipelines based on conditions

Span Metrics connector — Generates request, error, and duration (RED) metrics from span data

Sum connector — Aggregates telemetry data into sum metrics across pipelines

How Components Fit Together

A service pipeline chains receivers, processors, and exporters



Extensions run alongside pipelines (health checks, auth, discovery). Connectors join two pipelines together, acting as an exporter for one and a receiver for the other.

```
service:
  pipelines:
    metrics:
      receivers: [hostmetrics] processors: [batch] exporters: [signalfx]
```

Naming Convention: Snake Case Migration

Some multi-word component names are being renamed upstream

Old Name	New Name
azureblob	azure_blob
azureeventhub	azure_event_hub
azuremonitor	azure_monitor
filelog	file_log
httpcheck	http_check
mongodbatlas	mongodb_atlas
prometheusremotewrite	prometheus_remote_write
tcplog	tcp_log
udplog	udp_log
windowseventlog	windows_event_log

Next Steps

Explore components further

1

Process Your Data with Pipelines

Learn how to define and chain receivers, processors, and exporters.

2

Common Configuration Options

Review authentication, gRPC, HTTP, network, and TLS settings shared across components.

3

Use Tags or Attributes in OpenTelemetry

Understand how metadata flows through the pipeline via attributes.

4

`opentelemetry-collector-contrib`

Browse the upstream GitHub repository for the full list of available components.